2nd Year Computer Science — Core Concepts Study Guide

This reference document contains core theoretical concepts for Class 12 / 2nd Year Computer Science (Object-Oriented Programming and Database Systems) structured for document retrieval and indexing.

1. Object-Oriented Programming (OOP) in C++

1.1 Classes and Objects

- **Class:** A user-defined data type that acts as a blueprint or template for creating objects. It encapsulates data members (variables) and member functions (methods).
- **Object:** An instance of a class. Memory is allocated when an object is created, not when the class is defined.
- **Constructor:** A special member function that initializes objects of its class. It has the same name as the class and no return type.
- **Destructor:** A member function that is automatically called when an object is destroyed to release allocated memory.

1.2 Four Pillars of OOP

- **Encapsulation:** The bundling of data and functions operating on that data into a single unit (class), restricting direct access from outside the class.
- **Abstraction:** Hiding internal implementation details and showing only essential features to the user.
- **Inheritance:** The mechanism by which a derived (child) class inherits properties and behaviors from a base (parent) class, promoting code reusability.
- **Polymorphism:** The ability of a message or function to be displayed in more than one form (e.g., function overloading and runtime overriding).

2. Database Management Systems (DBMS) & SQL

2.1 Relational Database Fundamentals

- **Database:** An organized collection of structured data stored electronically for easy access, management, and updating.
- **Table (Relation):** A collection of data organized into rows (tuples/records) and columns (attributes/fields).
- **Primary Key:** A field or set of fields that uniquely identifies each record in a database table. It cannot contain duplicate or NULL values.
- **Foreign Key:** A column in one table that links to the Primary Key of another table, establishing a relationship between them.

2.2 SQL (Structured Query Language) Statements

- **Data Definition Language (DDL):** Commands like `CREATE`, `ALTER`, and `DROP` used to define and modify database structures.
- **Data Manipulation Language (DML):** Commands like `SELECT`, `INSERT`, `UPDATE`, and `DELETE` used to manipulate data stored in tables.